INTERVIEW PREP SERIES

HTML

Interview Questions & Easy Answers

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51 questions · 16 marked ★ most asked

Covers document structure, semantic tags, forms, media, accessibility, SEO, HTML5 APIs and the "what happens when..." questions asked in frontend and full stack rounds.

How to use this document

- Questions go from basic to advanced inside each section. Freshers should master the ★ ones first.
- Answers are written in simple words, the way you would say them in an interview. Learn the idea, not the exact sentences.
- Questions are based on technical rounds reported by candidates on GeeksforGeeks, LeetCode Discuss, Glassdoor, Naukri and similar interview-experience pages, plus the standard question banks interviewers draw from.
- Read one section a day and say each answer out loud once. Speaking it is what makes it stick.

Basics

Q1. What is HTML? ★ Most asked

HTML (HyperText Markup Language) is the standard language used to create the structure of web pages. It uses tags to tell the browser what each piece of content is: a heading, a paragraph, an image, a link and so on. HTML gives structure, CSS gives style, and JavaScript gives behaviour.

Q2. What is the basic structure of an HTML document? ★ Most asked

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Page</title>
  </head>
  <body>
    <h1>Hello</h1>
  </body>
</html>
```

`<head>` holds information about the page (title, meta, CSS links). `<body>` holds what users see.

Q3. What is `<!DOCTYPE html>`? ★ Most asked

It tells the browser that the page uses HTML5, so the browser renders it in "standards mode". Without it, the browser may switch to "quirks mode" and display things inconsistently. It is not an HTML tag, it is an instruction.

Q4. What is the difference between a tag and an element?

A tag is the code written in angle brackets, like `<p>` and `</p>`. An element is the complete thing: opening tag + content + closing tag, e.g. `<p>Hello</p>`.

Q5. What are attributes?

Attributes give extra information about an element and are written inside the opening tag as name="value". Examples: `href` in `<a>`, `src` and `alt` in `<img>`, `id`, `class`, `type`.

Q6. What are void (self-closing) elements?

Elements that have no content and no closing tag, like `<br>`, `<hr>`, `<img>`, `<input>`, `<meta>` and `<link>`.

Q7. What is the difference between `id` and `class`? ★ Most asked

An `id` must be unique on a page and identifies one element (`#header` in CSS). A `class` can be used on many elements to group them for styling (`.card` in CSS). One element can have many classes but only one id.

Q8. What are the heading tags?

`<h1>` to `<h6>`. `<h1>` is the most important (usually one per page, the main title) and `<h6>` the least. Search engines and screen readers use headings to understand the page outline, so do not skip levels just for size.

Q9. What is the difference between block-level and inline elements? ★ Most asked

Block elements start on a new line and take the full width available (`<div>`, `<p>`, `<h1>`, `<ul>`, `<section>`). Inline elements stay in the same line and take only the width of their content (`<span>`, `<a>`, `<strong>`, `<img>`).

Q10. What is the difference between `<div>` and `<span>`?

Both are generic containers with no meaning. `<div>` is block-level and used to group sections. `<span>` is inline and used to style a small part of text.

Semantic HTML

Q11. What is semantic HTML? Why is it important? ★ Most asked

Semantic tags describe the meaning of their content, not just how it looks. Examples: `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`, `<figure>`.

Benefits: better SEO (search engines understand the page), better accessibility (screen readers can navigate), and cleaner, easier-to-read code.

Q12. What is the difference between `<section>` and `<article>`?

`<article>` is self-contained content that makes sense on its own, like a blog post or a product card. `<section>` groups related content under a theme, usually with a heading, like a "Features" section of a page.

Q13. What is the difference between `<b>` and `<strong>`, `<i>` and `<em>`?

`<b>` and `<i>` only change looks (bold, italic). `<strong>` means "important" and `<em>` means "emphasis"; screen readers may stress them. Prefer `<strong>` and `<em>` for meaning.

Q14. What is the use of `<main>`?

`<main>` wraps the main unique content of the page, excluding the header, nav and footer. There should be only one visible `<main>` per page. It helps screen-reader users jump straight to the content.

Links, lists, tables and media

Q15. How do you create a link that opens in a new tab? ★ Most asked

```
<a href="https://example.com" target="_blank" rel="noopener noreferrer">Visit</a>
```

`target="_blank"` opens a new tab; `rel="noopener noreferrer"` is a security best practice so the new page cannot control your page.

Q16. What are the types of lists in HTML?

- Ordered list `<ol>` (numbered).
- Unordered list `<ul>` (bullets).
- Description list `<dl>` with `<dt>` (term) and `<dd>` (description).

Each item in `<ol>`/`<ul>` is an `<li>`.

Q17. How do you create a table?

```
<table>
  <thead><tr><th>Name</th><th>Age</th></tr></thead>
  <tbody><tr><td>Riya</td><td>22</td></tr></tbody>
```

```
</table>
```

`<tr>` = row, `<th>` = header cell, `<td>` = data cell. `colspan` and `rowspan` merge cells.

Q18. Why is the `alt` attribute important on images? ★ Most asked

`alt` gives a text description of the image. It is read by screen readers, shown if the image fails to load, and used by search engines. Decorative images should have `alt=""`.

Q19. How do you add audio and video in HTML5?

```
<video src="movie.mp4" controls width="400"></video>
<audio src="song.mp3" controls></audio>
```

`controls` shows play/pause buttons. You can add `autoplay`, `muted`, `loop` and multiple `<source>` tags for different formats.

Q20. What is an `<iframe>`?

An iframe embeds another web page inside your page, like a YouTube video or a Google Map. It should be used carefully, since loading other sites can affect security and speed.

Q21. What is the difference between `<img>` and `<picture>`?

`<img>` shows a single image. `<picture>` lets you give multiple `<source>` options so the browser picks the best one by screen size or format (like WebP), with `<img>` as a fallback.

Q22. What is `srcset`?

`srcset` lets you list several versions of an image at different sizes. The browser picks the right one for the screen, so mobile users download smaller files.

Forms

Q23. How do you create a form in HTML? ★ Most asked

```
<form action="/signup" method="POST">
  <label for="email">Email</label>
  <input type="email" id="email" name="email" required>
  <button type="submit">Sign up</button>
</form>
```

`action` is where data is sent, `method` is GET or POST, and `name` is the key the server receives.

Q24. What is the difference between GET and POST in forms? ★ Most asked

GET sends data in the URL (visible, limited length, can be bookmarked; good for search). POST sends data in the request body (not shown in URL, no size limit; used for login, signup, uploads).

Q25. What are some HTML5 input types?

`email`, `password`, `number`, `tel`, `url`, `date`, `time`, `color`, `range`, `file`, `checkbox`, `radio`, `search`. They give built-in validation and the right mobile keyboard.

Q26. What are form validation attributes?

`required`, `minlength`, `maxlength`, `min`, `max`, `pattern` (regex) and `type="email"`. The browser checks these before submitting. Always validate again on the server, since users can bypass browser checks.

Q27. Why use the `<label>` tag?

A label describes an input. Linking it with `for="id"` lets users click the label to focus the input, and screen readers announce it. It improves accessibility and usability.

Q28. What is the difference between `readonly` and `disabled`?

`readonly` inputs cannot be edited but are still submitted with the form and can be focused. `disabled` inputs cannot be edited, cannot be focused and are NOT submitted.

Q29. What is the difference between radio buttons and checkboxes?

Radio buttons (same `name`) allow only one choice from a group. Checkboxes allow selecting zero, one or many options.

Q30. What are `<select>`, `<textarea>` and `<datalist>`?

`<select>` makes a dropdown with `<option>`s. `<textarea>` is a multi-line text box. `<datalist>` gives suggestions for an `<input>` while still allowing free typing.

Head, meta and SEO

Q31. What are meta tags? ★ Most asked

Meta tags in `<head>` give information about the page to browsers and search engines. Common ones:

- `<meta charset="UTF-8">`: character encoding.
- `<meta name="viewport" content="width=device-width, initial-scale=1.0">`: makes the page responsive on mobile.
- `<meta name="description" content="...">`: summary shown in search results.

Q32. What is the viewport meta tag and why is it needed? ★ Most asked

Without it, mobile browsers show the page as a zoomed-out desktop page. `width=device-width, initial-scale=1.0` tells the browser to use the device's real width, which is required for responsive design.

Q33. How does HTML help with SEO?

Use a clear `<title>` and meta description, one `<h1>`, proper heading order, semantic tags, `alt` text on images, meaningful link text, and fast-loading pages. Open Graph meta tags control how links look when shared on social media.

Q34. What is the difference between `async` and `defer` on script tags? ★ Most asked

Both download the script without blocking HTML parsing.

- `async` runs the script as soon as it downloads, in any order. Good for independent scripts like analytics.
- `defer` runs scripts after HTML parsing is complete, in order. Good for your main app scripts.

A normal `<script>` blocks parsing while it downloads and runs, which is why scripts were often placed at the end of `<body>`.

Q35. Where should you put `<link>` (CSS) and `<script>` tags?

CSS `<link>` goes in the `<head>` so the page renders styled from the start. Scripts go at the end of `<body>` or in the `<head>` with `defer`.

HTML5 features and APIs

Q36. What are the new features in HTML5? ★ Most asked

Semantic tags, `<audio>` and `<video>`, `<canvas>` and SVG support, new input types and validation, `localStorage`/`sessionStorage`, Geolocation API, drag and drop, Web Workers, and `data-*` attributes.

Q37. What are `data-*` attributes?

Custom attributes to store extra data on an element, which JS can read through `dataset`.

```
<button data-user-id="42">View</button>
// JS: button.dataset.userId  -> "42"
```

Q38. What is the difference between `<canvas>` and `<svg>`?

`<canvas>` draws pixels using JavaScript, so it is fast for games and many moving objects, but the drawing is not a set of elements. SVG is vector, made of DOM elements, scales without blurring, and is good for icons, logos and charts.

Q39. What is Web Storage?

`localStorage` and `sessionStorage` let you store key-value strings in the browser. `localStorage.setItem("theme", "dark")`, `localStorage.getItem("theme")`. localStorage stays after closing; sessionStorage clears when the tab closes.

Q40. What are Web Workers?

Web Workers run JavaScript in a background thread so heavy work does not freeze the page. They cannot access the DOM directly and talk to the main page using messages.

Q41. What is the Geolocation API?

`navigator.geolocation.getCurrentPosition(success, error)` gets the user's location after they give permission. It only works on secure (HTTPS) pages.

Accessibility and browser

Q42. What is web accessibility (a11y)? ★ Most asked

Making websites usable by everyone, including people with disabilities. Key practices: semantic HTML, `alt` text, labels for inputs, keyboard navigation, enough colour contrast, and ARIA attributes when needed.

Q43. What are ARIA attributes?

ARIA (Accessible Rich Internet Applications) attributes add extra meaning for screen readers, like `aria-label`, `aria-hidden`, `aria-expanded` and `role="button"`. Rule: use native HTML elements first and ARIA only when HTML cannot do it.

Q44. What is `tabindex`?

`tabindex` controls keyboard focus order. `tabindex="0"` makes an element focusable in normal order, `tabindex="-1"` makes it focusable only by JS, and positive values should be avoided.

Q45. What happens when you type a URL in the browser and press Enter? ★ Most asked

1. The browser finds the IP address of the domain using DNS.

2. It opens a TCP connection (and a TLS handshake for HTTPS).

3. It sends an HTTP request.

4. The server sends back the HTML.

5. The browser parses HTML into the DOM and CSS into the CSSOM, combines them into a render tree, calculates layout, and paints pixels on the screen. Scripts and images are loaded along the way.

Interview tip: This is asked in almost every web interview. Mention DNS, TCP, HTTP, and DOM + CSSOM → render.

Q46. What is the critical rendering path?

The steps the browser takes to show a page: HTML → DOM, CSS → CSSOM, DOM + CSSOM → render tree, layout (positions and sizes), and paint. Making CSS small and deferring scripts makes this path faster.

Q47. What is the difference between HTML and XHTML?

XHTML is a stricter, XML-based version of HTML: all tags must be closed, lowercase, and properly nested. HTML5 is more forgiving and is what everyone uses today.

Q48. What is the difference between HTML and HTML5?

HTML5 is the latest version. It added semantic tags, native audio/video, canvas, new form inputs, local storage and many JS APIs, and it uses the simple `<!DOCTYPE html>`.

Q49. What are HTML entities?

Special codes to display reserved characters: `<` for <, `>` for >, `&` for &, ` ` for a non-breaking space, `©` for ©.

Q50. What is lazy loading of images?

Adding `loading="lazy"` to `<img>` or `<iframe>` tells the browser to load it only when it comes near the screen. It makes pages load faster and saves data.

Q51. What is the difference between `<link>` and `<a>`?

`<link>` lives in `<head>` and connects a resource like a stylesheet or favicon; users do not click it. `<a>` is a clickable hyperlink in the page body.